Deepfakes: A Review of Creation and Research Trends

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Abstract. Over the past few years, deepfake technology has become significantly widespread, driven by advances in deep learning models that create artificial images, videos, and audio. These developments raise serious concerns regarding the trust, privacy and security of the public. Although deepfakes are often used in various sectors, such as entertainment, education, and access, they are equally often used to commit malicious acts, such as creating non-consensual content, sharing political misinformation, committing identity theft, and committing financial fraud. With the introduction of advanced deep learning networks, the creation of such synthetic media has become democratised; therefore, the process of detection has become considerably more challenging. This paper presents an overview of deepfake technologies by explaining their fundamental concepts, generation methodologies, primary challenges associated with deepfake generation, social impacts, and existing detection and identification techniques. In addition, available benchmark datasets and tools, and also trend analysis of research publications, highlight the rapid evolution of the field. This survey serves as an accessible entry point for researchers and non-experts to understand deepfake-related concepts, methods, and emerging research impacts and trends.

Keywords: Deep Learning (DL) · Artificial Intelligence (AI) · Deepfake · Synthetic Media · Deepfake Detection · Cybersecurity

1 Introduction

Computer-generated imagery (CGI) is a growing field that has enabled manipulation and animation of visual media in the 1990s and 2000s. These primitive technologies have evolved over the years to produce modern artificial intelligence systems that can produce synthetic media [5]. The term *'deepfake'* first emerged in 2017, after a Reddit user published synthetic face manipulation content, and it was circulated across social media platforms. The name is a coinage from 'deep learning' and 'fake' media. Although such technologies have legal approaches

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and applications in usage areas such as entertainment, education, and accessibility, instead they are frequently applied for harmful or unethical purposes. Severe abuse cases include the generation of illegal materials, the propagation of politically inclined misinformation, the impersonation of identity-based fraud, reputational damage, and financial fraud. These practices are a significant risk to the trust of the public, digital privacy, and the security of information [31, 39, 40].

The studies of generative neural network architecture with high computational capability have significantly improved the realism of synthetically produced media. These improvements make it harder to detect synthetic content using both human perception and automated systems for its detection due to the fact that such advancements make it more challenging [34, 39]. To counter such risks, governments and technology organisations have started to place policy guidelines, regulatory advisories, and countermeasures to combat and reduce the risk involved [8, 23].

2 Deepfake Technology: Concepts and Societal Impact

2.1 Deepfake Types and Creation Methods

Deepfakes are AI-generated media in which learning models are used to manipulate facial identity, expressions, or speech to imitate real individuals with high visual or auditory realism. Face swapping, voice cloning, lip-synchronisation editing, and full facial synthesis are conventional forms of manipulation modalities, and they generate artefacts of entirely artificial facial representations without their concerns or direct connection to an individual [19, 22, 34]. Several generative modelling methods are explained as follows:

Generative Adversarial Networks (GANs): GAN-based systems have two adversarial neural networks working together, where the first neural network named as generator produces synthetic samples while the other named as discriminator evaluates whether the content produced by it appears authentic. Though iterative, competition between these two networks over time is capable of producing progressively realistic visual outputs by the first neural network.

Autoencoder-based architectures: Autoencoders have been actively used in face reenactment and identity swapping. These models learn compact latent representations of facial features, which can then be decoded to reconstruct or modify facial identities in target media [16].

Diffusion models: Diffusion models are different kinds of content-generating models that learn a denoising trajectory to generate high-fidelity structural visual artefacts by conditioning on progressively increasing amounts of stochastic noise. Diffusion-based methods tend to achieve greater levels of fidelity and have few obvious artefacts compared to past generations of generative paradigms [14]. These models are moderated by the size of training datasets, complexity of the model and available computational resources, which have the most significant role in determining the quality of the outputs of these models [5, 35].

2.2 Challenges, Societal Impact and Regulations

Privacy and Sexual Abuse: The most commonly reported abuse is the creation of non-consensual explicit content, where the faces of victims are overlaid onto the fabricated media, thus causing them emotional disorders, social stigma and damage to their reputations [33]. **Political Misinformation:** Deepfake videos of political leaders can propagate misinformation, manipulate public opinion, and influence electoral outcomes, which can bring social instability. Moreover, research suggests that the psychological impact on public opinion can still be experienced even after the content is deleted [40]. **Financial Fraud and Identity Impersonation:** Financial frauds use deepfake audio and video to impersonate corporate executives or officials, or even family members, to deceive victims by transferring funds or giving confidential data, posing a severe threat to cybersecurity and global economic stability [8, 33].

Deepfake technology has become a critical international issue, which affects digital trust, cybersecurity, political stability, and informational integrity. Several global organisations have reported a rapid increase in the abuse of synthetic media across social, political and economic segments [40]. In response, the governments and tech giants around the global have initiated collaborative efforts to develop detection frameworks, such as the Coalition for Content Provenance and Authenticity (C2PA) and innovative blockchain-based architectures for media verification [2]. Deepfake content online has grown over 550% since 2019, and approximately 98% of deepfake content found online is related to non-consensual explicit content [33]. Cybersecurity using deepfakes have risen over 1,300% annually [10]. The deepfake market is estimated at approximately USD 805 million in 2023 and is expected to grow by USD 6.22 billion by 2033 [12].

In India, the Ministry of Electronics and Information Technology (MeitY) issued several rules to social media firms for blocking the deepfake content in accordance with the IT Rules, 2021 [23]. Additionally, the Indian Computer Emergency Response Team (CERT-In) is recommending countermeasures such as multi-factor authentication and public awareness initiatives [3], and also regulated measures that are reinforced by the suggested amendments to the IT Rules in 2025, which will require that synthetic media be labelled and that platforms spreading them be held fully accountable for such content [24].

3 Deepfake Detection Techniques and Identification

Similarly, rapid advancements in deepfake generation have created significant issues for digital forensics, multimedia security, and cybersecurity research in these areas of technology [28]. Several detection methods are described as follows:

Conventional Forensic-Based Detection: Initial research on deepfake detection methods was based on the field of digital media forensics, which led to detecting visual anomalies that occur during manipulation [35]. These techniques examined abnormal face boundaries, lighting inconsistencies, asymmetries in shadows, merging artefacts and abnormal eye-blink patterns [17]. A different research question explored physiological signals, including remote photoplethysmography (rPPG), that measures changes in the skin colour of blood flowing under the skin. Since such biological patterns are often difficult to fit into synthetic videos, an analysis of rPPG can reveal manipulated content [17]. Though computationally efficient, these methods have eventually lost their strength to reliability, with modern generative models eliminating apparent artefacts of the output.

Deep Learning-Based Detection Methods: CNN-based methods are widely used to learn spatial patterns that distinguish real content from manipulated ones using large training datasets with the intention of automatic learning tasks [20, 32]. Transformer-based architectures enhance the performance of detection by modelling global contextual dependencies across image regions and video frames [30]. Temporal consistency between adjacent frames is a key feature in video-based detection because distinctions in videos are usually subtle temporal inconsistencies. RNNs, LSTM networks, and 3D-CNNs are the sequential learning models that examine frame-to-frame relationships to recognise abnormal patterns added to the synthesis process [13].

Furthermore, the main drawback of the methods is that they are prone to overfitting to manipulation strategies that are present during training and end up in suboptimal generalisation between unseen deepfake techniques and cross-dataset environments [32, 34].

Audio-Based and Multimodal Detection: Recent studies have demonstrated that when multiple modalities are combined, deepfake detection performance is improved significantly. Audio-based methods examine speech characteristics, including variations in pitch level, spectral distribution, prosody, and vocal dynamics, in order to determine patterns of synthetic speech [1, 19]. Multimodal detection systems are able to simultaneously analyse visual features, audio-video synchronisation, and semantic alignment across streams. The analysis of cross-modal inconsistency helps achieve better results in the performance of detection against sophisticated audiovisual deepfakes [19, 36].

3.1 Generalisation, Robustness, and Content Provenance

A major limitation of the above detection methods is that they have low cross-domain generalisation; the models are frequently trained on one dataset and then become irrelevant when faced with unfamiliar manipulation strategies or real-world distributions to detect them [34]. Other artefacts such as compression, adversarial perturbations, and diffusion-based synthesis also degrade detector performance even further [6]. Another paradigm is based on authenticity and is followed through media provenance, whereby at the time of creation, the content is signed with cryptographic signatures, metadata logs or invisible watermarks [2], hence allowing verification without artefact detection.

4 Deepfake Datasets and Tools

Deepfake generation and detection technique research heavily relies on large-scale datasets for training, validation and performance evaluation using different architectures. The datasets are mentioned in Table 1.

Table 1: Deepfake benchmark datasets are categorised by modality, scale, and citation impact (Google Scholar).

Dataset	**Year**	**Modality**	**Scale**	**Citations**
FFHQ [18]	2019	Image	70,000 images	∼17,184
CelebA-Spoof [44]	2020	Image	625,537 images	∼313
DIRE [38]	2023	Image	615,200 images	∼528
Semi-Truths [29]	2024	Image	1,723,700 images	∼10
FaceForensics++ [32]	2019	Video	5,000 videos	∼4,049
Celeb-DF v2 [22]	2020	Video	6,229 videos	∼2,316
DFDC (Full) [9]	2020	Video	128,154 videos	∼1,484
DeeperForensics-1.0 [15]	2020	Video	60,000 videos	∼754
KoDF [21]	2021	Video	37,942 videos	∼204
ASVspoof2019 (LA task) [37]	2019	Audio	121,461 utt.	∼662
ASVspoof2021 (LA task) [41]	2021	Audio	164,640 utt.	∼582
WaveFake [11]	2021	Audio	117,000 utt.	∼244
ADD 2022 [42]	2022	Audio	154,949 samples	∼298
ADD 2023 [43]	2023	Audio	517,068 utt.	∼197
MLAAD [27]	2024	Audio	163.9 hours	∼135
FakeAVCeleb [19]	2021	A+V	20,000 videos(Fake+Real)	∼410
Deepfake-Eval-2024 [4]	2024	A+V+I	45 hrs video, 56.5 hrs audio, 1,975 images	∼41
InDeepFake [7]	2025	A+V	5,069 videos(Fake+Real)	∼3
DDL [25]	2025	A+V+I	1,400,000+ samp.	∼2

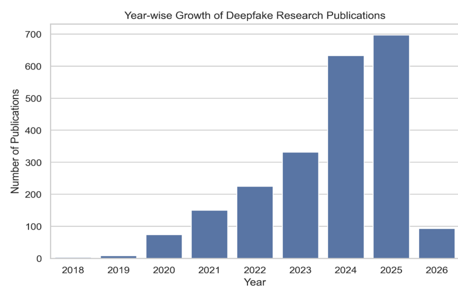
utt. = utterances; A+V = audio-visual; A+V+I = audio, video & image;

Available tools related to different modalities: The generation of images, such as StyleGAN2, Stable Diffusion, DALL·E, Midjourney and Roop, can be seen as a continuum between research-grade and face-swap applications easy to find on the web. Video generating tools include DeepFaceLab, FaceSwap, SimSwap, First Order Motion Model and Wav2Lip. Audio synthesis tools include both commercial voice-cloning services like ElevenLabs and Resemble AI and open-source systems like SV2TTS, Coqui TTS and Descript [26, 28].

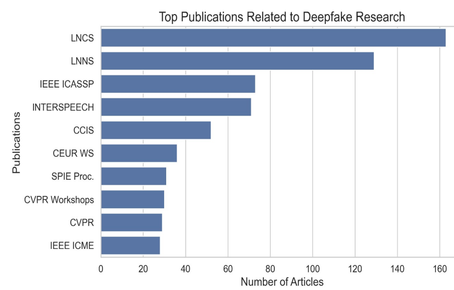
However, most of the existing datasets are typified by demographic imbalance and low manipulation diversity, which leads to systematic bias in the corresponding detection models and their inability to generalise the authentic deepfake scenarios [26, 34].

5 Bibliometric Visualisation and Research Trends

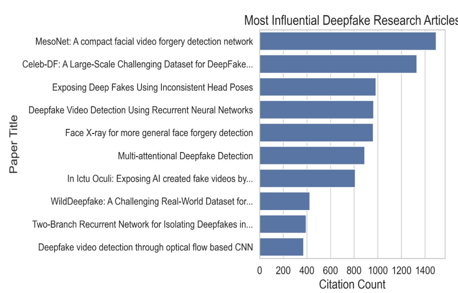
A total of 2,214 English-language research papers were retrieved from Scopus using the keyword 'deepfake', ranging from 2018 to January 2026.



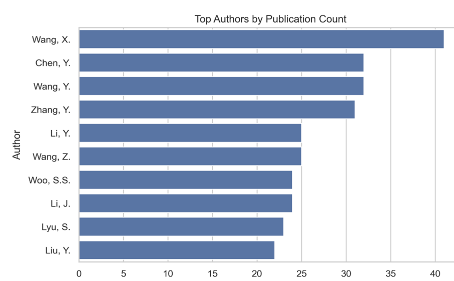
(a) Year-wise publication growth (2018–2026).



(b) Deepfake-related publications across venues.



(c) Most influential papers by citation count.



(d) Top Authors by publication count.

Fig. 1: Bibliometric analysis of deepfake research (Scopus, 2018–2026).

Fig 1a shows publications were negligible until 2019 (∼8 papers), after which a sharp increase is observed, rising from ∼75 in 2020 to ∼700 in 2025. This growth coincides with the emergence of large-scale benchmark datasets, open-source tools, advancements in GAN-based generation techniques, and growing social concerns over synthetic media. Publication venues reflect the interdisciplinary nature of this growth, as shown in Fig. 1b. Lecture Notes in Computer Science (LNCS, ∼163 articles) and Lecture Notes in Networks and Systems (LNNS, ∼130) dominate, while the strong presence of IEEE ICASSP (∼73) and INTERSPEECH (∼71) highlights significant deepfake literature.

The most cited papers on Scopus are based on a bibliometric search as shown in Fig. 1c. MesoNet (∼1,450 citations) and Celeb-DF (∼1,350 citations), highly cited works, typically introduce benchmark datasets and foundational detection architectures that disproportionately draw scholarly attention and continue to anchor the field's evaluation standards. A middle tier of works on head-pose inconsistency, recurrent detection, and Face X-ray each contribute approximately 975 citations, while the remaining entries fall in the 375–825 range. The number of publications done by authors on Scopus based on a bibliometric search as shown in Fig. 1d. The top authors' publication count is led by Wang, X. (∼41 papers), followed by Chen, Y., and Wang, Y. (∼32 each), and Zhang, Y. (∼31). Remaining authors clustered in the range of 22 to 25 publications, indicating a strong but related core of active contributors shaping the field. Overall, the bibliometric-based trends confirm convergence towards robust detection of multimodal detection and scalable benchmark development are the most favoured research directions.

6 Conclusion, Limitations and Future Directions

This paper presents an overview of deepfake technology, covering generation methods, societal implications, detection techniques, benchmark datasets, and research trends. Analysis of detection methodologies shows that there has been a significant shift from traditional forensic methods towards deep learning-based, temporal, and multimodal verification strategies due to the growing complexity of audiovisual deepfake attacks and the inability of single-modality detectors to address them.

The present study has certain limitations: the bibliometric analysis is limited to English-language records and keyword-specific searching on the Scopus database, and citation-based metrics may introduce temporal bias favouring older publications. Existing datasets are technically limited by demographic imbalance, generation-method bias, and susceptibility to diffusion-based synthesis, which has outpaced the detection capabilities of many established models.

Future research should prioritise generalised detection for diffusion-based synthetic media and multimodal manipulation, with diverse cross-lingual datasets and cross-sector collaboration to ensure responsible adoption of generative AI technologies.

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